
Beyond the Transcript: Reliability and Bias in LLM-Based Depression Judgments

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Abstract

LLM-as-judge systems are increasingly used to automate subjective evaluation [1, 3, 7], but in high-stakes settings it is unclear whether their judgments reflect the intended construct or artifacts of the measurement procedure. We study this question for depression screening from clinical dialogue using DAIC-WOZ. We evaluate PHQ-8-inspired judgments from GPT-4o, Claude Sonnet 4.6, and Gemini 2.5 Pro across participant-only, full-transcript, and interviewer-only conditions, comparing structured item-threshold scoring with direct global binary judgments. Across participant-containing conditions, global judgments achieve higher recall, F1, MCC, cross-model reliability, and dependability than item-threshold predictions. Speaker ablations show that interviewer-only inputs are unreliable overall but reveal model-specific shortcut sensitivity: Claude remains highly predictive from interviewer-only text, whereas GPT-4o and Gemini largely do not. Gender-metadata perturbations show that actual-gender metadata rarely changes global binary judgments, though item-threshold judgments are more sensitive. These results suggest that participant speech provides most of the usable signal, but zero-shot LLM depression judgments are still shaped by output format, model identity, and, for some models, interviewer-side artifacts. Overall, LLM judges may provide preliminary screening signals, but input ablations, cross-model reliability checks, and demographic metadata sensitivity tests are necessary before any high-stakes use.

1 Introduction and Background

A useful measurement instrument should respond to the construct it is intended to measure, not to irrelevant features of the measurement procedure. For LLM-based depression screening, this distinction is especially important. A model may appear to judge depression severity from a clinical interview transcript, but its output could also be influenced by speaker role, scripted interviewer prompts, output format, model-specific judging behavior, or demographic metadata.

This project sits at the intersection of two research areas. First, LLM-as-judge systems are used as automated evaluators in settings where human annotation is expensive, subjective, or difficult to scale, including dialogue, instruction-following, and preference evaluation. Second, clinical NLP has studied mental-health screening from interview and dialogue data, including DAIC-WOZ depression assessment [2, 5]. Combining these areas raises a high-stakes measurement question: when an LLM judge reads a clinical interview transcript, does it mainly measure participant symptom evidence, or does it respond to artifacts of the interview format?

This concern is especially important for semi-structured clinical interviews. In DAIC-WOZ, participants interact with a virtual interviewer, Ellie, whose prompts are drawn from a controlled repertoire designed to make interviews consistent across participants [2]. While this structure supports standardized data collection, it can also introduce shortcut cues. Watawana et al. [6] show that supervised

depression classifiers can exploit interviewer prompts and prompt positions: interviewer-only models can match or outperform participant-only models across DAIC-WOZ, E-DAIC, and ANDROIDS. This finding motivates our study. We ask whether a similar speaker-side shortcut appears when the supervised classifier is replaced by a zero-shot LLM judge.

Our main research question is:

Do zero-shot LLM judges produce depression judgments that mainly reflect participant speech, or are their judgments sensitive to interview structure, output format, model identity, and demographic metadata?

We study three related questions. First, if participant speech contains most of the relevant symptom evidence, participant-only and full-transcript inputs should produce broadly similar judgments, while interviewer-only inputs should perform poorly or inconsistently. Second, if the task is well-defined as a measurement procedure, GPT-4o, Claude, and Gemini should tend to agree when judging the same interview under the same condition. Third, adding participant gender metadata to the same transcript should not substantially change binary outputs. This final analysis is not a complete fairness audit; rather, it is a metadata-sensitivity check motivated by the principle that the same participant evidence should not receive a different judgment solely because a demographic label is appended to the prompt.

We answer these questions using speaker ablations, cross-model reliability statistics, Generalizability Theory dependability analysis, many-facet Rasch-style modeling, and demographic metadata perturbations. Code for the analyses is available at <https://github.com/kerui-lu/CS321M-Project>.

2 Methods

2.1 Dataset and Input Conditions

We use the DAIC-WOZ development split, which contains $n = 35$ interviews with available PHQ-8 total scores and binary depression labels. Twelve of the 35 interviews are depression-positive, defined as $\text{PHQ8_Binary} = 1$. PHQ-8 is a standard self-report screening measure for current depression symptoms [4].

Each interview is evaluated under three input conditions:

- **Participant-only (P):** only participant utterances are provided.
- **Interviewer-only (I):** only interviewer utterances are provided.
- **Full transcript (F):** both participant and interviewer utterances are provided.

For the gender-metadata sensitivity analysis, we compare no-gender and actual-gender variants for both the participant-only and full-transcript conditions. In the no-gender variant, the transcript is provided without explicit participant gender metadata. In the actual-gender variant, the same transcript is paired with the participant’s recorded gender. We do not run the gender-metadata comparison for interviewer-only because that condition removes the participant language we want to hold fixed. Adding gender metadata to interviewer-only text would primarily test whether a demographic label interacts with interview structure, rather than whether the same participant evidence is judged differently.

2.2 LLM Judges and Output Formats

We evaluate three LLM models: GPT-4o, Claude Sonnet 4.6, and Gemini 2.5 Pro. For each input condition, each model produces two outputs.

The first is a structured PHQ-8-inspired item-evidence score. We sum the eight predicted symptom-item scores and threshold the total at 10:

$$\hat{y}_{\text{item}} = \mathbf{1}[\text{pred_evidence_total} \geq 10].$$

We call this the **item-threshold** method. It approximates a structured symptom-evidence workflow, but is not equivalent to administering the PHQ-8 questionnaire.

Table 1: Model-level comparison on participant-containing conditions.

Model	Method	Balanced accuracy	Recall	F1	MCC
GPT-4o	Item-threshold	0.667	0.333	0.500	0.497
GPT-4o	Global	0.793	1.000	0.716	0.572
Claude	Item-threshold	0.573	0.146	0.255	0.318
Claude	Global	0.784	0.917	0.710	0.542
Gemini	Item-threshold	0.665	0.396	0.521	0.410
Gemini	Global	0.746	0.917	0.672	0.476

Table 2: Input ablation without gender metadata for the global method. “Pooled” combines outputs from all three judge models, yielding $35 \times 3 = 105$ judgments per pooled row.

Input	Model	Pred+	Bal. acc.	Recall	Spec.	F1	MCC
Full transcript (F)	Pooled	59	0.791	0.944	0.638	0.716	0.557
Participant-only (P)	Pooled	62	0.769	0.944	0.594	0.694	0.520
Interviewer-only (I)	Pooled	20	0.588	0.306	0.870	0.393	0.212
Interviewer-only (I)	GPT-4o	1	0.478	0.000	0.957	NA	-0.124
Interviewer-only (I)	Claude	17	0.828	0.917	0.739	0.759	0.623
Interviewer-only (I)	Gemini	2	0.457	0.000	0.913	NA	-0.178

The second is a direct holistic binary judgment, which we call the **global** method. Here, the model directly predicts whether the transcript should be labeled depression-positive. Comparing these formats lets us test whether structured item accumulation or holistic judgment is more reliable in the zero-shot setting.

3 Results and Discussion

3.1 Global Judgments Perform Better than Item-Threshold Predictions

Table 1 compares the two output formats across models. We omit raw accuracy because the development split is class-imbalanced.

The item-threshold method is conservative and performs worse than the global method on all reported metrics. For all three models, item-threshold predictions have lower balanced accuracy, recall, F1, and MCC than global binary predictions. The largest gap is in recall: GPT-4o reaches only 0.333, Claude 0.146, and Gemini 0.396 under item-threshold predictions, compared with 1.000, 0.917, and 0.917 under the global method. This indicates that item-threshold predictions misses many depression-positive interviews, which also lowers F1 and MCC. In contrast, the global method captures more depression-positive interviews and therefore achieves stronger F1 and MCC across all three models. This pattern suggests that direct global predictions are better aligned with the binary label in this zero-shot setting, while item-threshold predictions behaves as a stricter and more conservative measurement procedure.

3.2 Input Ablation Shows Model-Specific Interviewer Sensitivity

Table 2 focuses on the global method, since Table 1 showed that it outperforms item-threshold predictions. Full-transcript and participant-only inputs have similar balanced accuracy (0.791 vs. 0.769) and identical recall (0.944), suggesting that participant speech is sufficient for most of the signal used by the zero-shot global judge. In contrast, pooled interviewer-only performance is much weaker: balanced accuracy drops to 0.588, F1 to 0.393, and MCC to 0.212.

The model-specific rows show why the interviewer-only condition remains important as a shortcut diagnostic. Claude reaches balanced accuracy 0.828, recall 0.917, and F1 0.759 using only interviewer utterances, while GPT-4o and Gemini have zero recall in the same condition. This partially echoes Watawana et al.’s finding that supervised interviewer-only classifiers can exploit fixed prompts

Table 3: Cross-model measurement reliability by condition. κ denotes average pairwise Cohen’s kappa across judge models, and α denotes Krippendorff’s alpha across GPT-4o, Claude Sonnet 4.6, and Gemini 2.5 Pro.

Condition	Item κ	Item α	Global κ	Global α	Evidence α
Full transcript (F)	0.543	0.534	0.845	0.847	0.860
Participant-only (P)	0.397	0.347	0.921	0.922	0.787
Full transcript + gender	0.575	0.592	0.845	0.846	0.859
Participant-only + gender	0.563	0.598	0.880	0.881	0.796
Interviewer-only (I)	NA	NA	0.047	-0.040	-0.122

and positions in semi-structured interviews [6], but the effect is not uniform in our zero-shot setting. Instead, shortcut sensitivity depends on both judge model and output format.

This distinction matters because our LLM judges are zero-shot rather than trained on the development split. Claude’s interviewer-only performance suggests that LLMs may still infer severity from follow-up patterns, scripted question order, or interviewer behavior. This supports speaker-aware ablations before interpreting full-transcript performance as participant-symptom measurement.

3.3 Global Judgments Have Higher Cross-Model Reliability

Table 3 evaluates inter-rater reliability across judge models. We treat GPT-4o, Claude Sonnet 4.6, and Gemini 2.5 Pro as independent raters scoring the same interviews. Average pairwise Cohen’s κ measures chance-corrected pairwise agreement, while Krippendorff’s α summarizes agreement across all three judge models.

For participant-containing conditions, global judgments show substantially higher inter-rater reliability across judge models than item-threshold predictions. The participant-only global condition has the strongest reliability, with $\kappa = 0.921$ and $\alpha = 0.922$. This indicates that the global output format defines a more stable binary measurement task across LLM raters.

The evidence-total α column measures reliability of the summed PHQ-8-style item-evidence score before thresholding. For participant-containing conditions, evidence-total α ranges from 0.787 to 0.860, suggesting that judge models often agree on the broader severity signal. However, converting this score into a binary label at the cutoff of 10 can amplify small differences near the threshold, reducing agreement on the final item-threshold decision.

The interviewer-only condition behaves as a shortcut diagnostic rather than a dependable measurement condition. Global reliability nearly disappears, with $\kappa = 0.047$ and $\alpha = -0.040$. The negative evidence-total α further suggests that judge models do not share a stable severity ordering when only interviewer utterances are provided. This reinforces that interviewer-only text should not be treated as a reliable source of participant-symptom measurement.

3.4 Generalizability Theory Dependability Analysis

To evaluate measurement reliability beyond predictive accuracy, we conduct a Generalizability Theory analysis of the binary depression judgments. The object of measurement is the interview I , and the measurement facets are judge model M and input condition C . We use a balanced, fully crossed $I \times M \times C$ design: every interview is evaluated by every judge model under every input condition. This allows us to decompose observed judgment variance into between-interview variance, judge-model variance, input-condition variance, their interactions, and residual error.

Because the object of measurement is the interview, between-interview variance σ_I^2 is the desired signal. Judge-model variance σ_M^2 and input-condition variance σ_C^2 are treated as measurement error for absolute decisions: a dependable depression judgment should not depend strongly on whether GPT-4o, Claude Sonnet 4.6, or Gemini 2.5 Pro produced the rating, or on which participant-containing variant was used.

We focus on the dependability coefficient Φ , which measures reliability for absolute decisions such as classifying an interview as depression-positive versus depression-negative. Unlike relative reliability,

which evaluates whether interviews are ranked consistently, Φ includes all sources of measurement error that can affect an absolute cut-score decision. We report whether Φ exceeds 0.80, a common heuristic for acceptable reliability in applied measurement settings. We use this threshold only as a reference point, not as a clinical validation standard. Because the outcome is binary, we interpret the results as a descriptive linear G-theory approximation to binary judgments.

For this analysis, C includes four participant-containing variants: full transcript, participant-only, full transcript with actual-gender metadata, and participant-only with actual-gender metadata. We exclude interviewer-only prompts because they are used as a shortcut diagnostic rather than as candidate measurement conditions. Thus, each G-study uses 35 interviews, 3 judge models, and 4 input conditions, yielding $35 \times 3 \times 4 = 420$ binary judgments per output format.

Table 4: G-study variance components for the item-threshold binary depression judgment.

Source	σ^2	% Total	Note
I : interview	0.04939	47.2	intended signal
M : judge model	0.00369	3.5	
C : input condition	0.00044	0.4	
$I \times M$	0.01139	10.9	
$I \times C$	0.00432	4.1	
$M \times C$	0.00000	0.0	clamped
$I \times M \times C + e$	0.03546	33.9	residual
Total	0.10469	100.0	

Item-threshold method. For the item-threshold method, Table 4 shows that only 47.2% of the variance is attributable to between-interview differences, which are the intended signal in this design. The remaining variance reflects measurement error for an absolute depression-positive decision. The large residual component and nontrivial $I \times M$ interaction indicate that item-threshold judgments vary substantially across judge models and idiosyncratic interview–model combinations.

Table 5: D-study dependability coefficients Φ for the item-threshold method. Values above 0.80 are bolded.

$n_M \backslash n_C$	$n_C = 1$	$n_C = 2$	$n_C = 3$	$n_C = 4$
$n_M = 1$	0.472	0.584	0.634	0.663
$n_M = 2$	0.622	0.724	0.767	0.790
$n_M = 3$	0.696	0.788	0.824	0.843

Table 5 reports the D-study dependability coefficient Φ for designs that average over different numbers of judge models and input conditions. For item-threshold judgments, a single judge model and single input condition yields $\Phi = 0.472$, well below the $\Phi \geq 0.80$ benchmark. The threshold is reached only when averaging over at least three judge models and three input conditions. This suggests that item-threshold judgments require substantial aggregation to support dependable absolute depression-positive decisions in this setting.

Global judgment method. For the global judgment method, the same $35 \times 3 \times 4$ fully crossed design yields $N = 420$ binary judgments. Table 6 reports the estimated variance components.

For the global judgment method, Table 6 shows that 86.3% of the variance is attributable to between-interview differences, the intended signal in this design. Judge-model and input-condition main effects are very small, and the residual component is much lower than for the item-threshold method. This indicates that global judgments are more consistently driven by interview-level differences rather than by the choice of judge model or participant-containing conditions.

Table 7 reports the D-study dependability coefficient Φ for global judgments under designs that average over different numbers of judge models and input conditions. The global method exceeds the $\Phi \geq 0.80$ benchmark even with one judge model and one input condition, where $\Phi = 0.863$. Averaging over additional judge models and conditions further improves dependability, but the marginal gains are modest because the single-measure design is already above the benchmark.

Table 6: G-study variance components for the global binary depression judgment.

Source	σ^2	% Total	Note
I : interview	0.21562	86.3	intended signal
M : judge model	0.00077	0.3	
C : input condition	0.00023	0.1	
$I \times M$	0.01669	6.7	
$I \times C$	0.00294	1.2	
$M \times C$	0.00000	0.0	
$I \times M \times C + e$	0.01349	5.4	residual
Total	0.24974	100.0	

Table 7: D-study dependability coefficients Φ for the global judgment method. Values above 0.80 are bolded.

$n_M \backslash n_C$	$n_C = 1$	$n_C = 2$	$n_C = 3$	$n_C = 4$
$n_M = 1$	0.863	0.893	0.904	0.909
$n_M = 2$	0.920	0.940	0.947	0.951
$n_M = 3$	0.941	0.957	0.963	0.965

Interpretation. The G-theory results support the same output-format pattern as the predictive-performance and inter-rater-reliability analyses. Item-threshold judgments have lower dependability and require aggregation over multiple judge models and input conditions to reach $\Phi \geq 0.80$. In contrast, global judgments exceed this benchmark even with one judge model and one input condition. The largest non-signal error component is the $I \times M$ interaction, indicating that judge models differ in how they classify particular interviews. This suggests that, when additional measurement budget is available, averaging over multiple judge models is likely more useful than adding more input-condition variants.

3.5 Many-Facet Rasch Analysis of Binary Depression Judgments

The G-theory analysis quantifies dependability, but it does not identify which facets systematically make judgments more or less severe. To examine these facet-level shifts, we fit a many-facet Rasch-style logistic model to the binary LLM depression judgments. The model treats each interview as the object of measurement and estimates whether judge model, input condition, or scoring format changes the log-odds of a depression-positive judgment after controlling for interview differences:

$$\text{logit}(\Pr(Y_{imcs} = 1)) = \theta_i + \alpha_m + \gamma_c + \delta_s,$$

where θ_i captures interview-level severity, α_m judge-model severity, γ_c input-condition effects, and δ_s scoring-format severity. Positive facet coefficients indicate a greater tendency to assign a depression-positive judgment, holding the other facets fixed.

We use this model descriptively, not as a calibrated diagnostic model. The goal is to compare facet effects: whether some judge models are more severe, whether some input conditions shift judgments, and whether item-threshold scoring is systematically stricter than global judgment. Estimates are log-odds coefficients; the 95% confidence interval (CI), standard error (SE), and p -value summarize uncertainty relative to the reference level.

Interpretation. The largest facet effect is scoring format. The item-threshold coefficient is large, negative, and statistically significant, indicating that item-threshold scoring is much more conservative than the global binary judgment after adjusting for interview, judge model, and input condition. This supports the earlier finding that item-threshold scoring behaves as a stricter measurement procedure.

Judge model also contributes systematic severity differences. Relative to Claude Sonnet 4.6, both Gemini 2.5 Pro and GPT-4o have positive coefficients, meaning that they are more likely to assign depression-positive judgments after controlling for interview, input condition, and scoring format. We interpret these coefficients as rater-severity differences, not as evidence of higher accuracy.

Table 8: Facet effects from the many-facet Rasch-style logistic model. Reference levels are Claude Sonnet 4.6, actual-gender full transcript, and global judgment. Positive estimates indicate a greater tendency to assign a depression-positive judgment.

Facet	Level	Est.	95% CI	SE	<i>p</i>
Model	Gemini	+3.331	[+1.934, +4.729]	0.713	< .001
Model	GPT-4o	+2.758	[+1.385, +4.131]	0.700	< .001
Condition	Gender + P	-0.311	[-1.407, +0.784]	0.559	0.578
Condition	No gender + F	-0.474	[-1.581, +0.633]	0.565	0.402
Condition	No gender + P	+0.295	[-0.772, +1.362]	0.544	0.588
Format	Item-threshold	-10.737	[-13.703, -7.771]	1.513	< .001

Table 9: Sensitivity to recorded gender metadata. Pred+ NG and Pred+ G are the numbers of positive judgments without and with recorded gender metadata. Δ values are actual-gender minus no-gender performance.

Input	Method	Pred+ NG	Pred+ G	Changed outputs	Bal. acc. Δ	F1 Δ	MCC Δ
Full transcript	Item	12	14	4/105	+0.028	+0.062	+0.050
Full transcript	Global	59	60	3/105	-0.007	-0.007	-0.012
Participant-only	Item	14	8	10/105	-0.041	-0.122	-0.044
Participant-only	Global	62	64	2/105	-0.014	-0.014	-0.024

In contrast, participant-containing input-condition effects are small and not statistically distinguishable from zero. After controlling for interview severity, judge model, and scoring format, participant-only versus full-transcript prompts and no-gender versus actual-gender variants do not produce large systematic shifts in binary judgments.

Overall, the many-facet Rasch analysis complements the G-theory results. G-theory shows that global judgments are more dependable, while the Rasch-style model shows that the main systematic facet effects are scoring format and judge model rather than participant-containing input condition.

3.6 Gender Metadata Sensitivity Is Small for Global Judgments

We next test whether adding recorded gender metadata changes model outputs when the transcript content is held fixed. This analysis compares no-gender and actual-gender prompt variants within the same transcript condition. Table 9 reports pooled changes across the three judge models. Each row contains $35 \times 3 = 105$ model-interview judgments. Changed outputs counts how often the binary judgment flips after adding recorded gender metadata, and Δ columns report actual-gender minus no-gender performance.

Global judgments are relatively stable to the addition of recorded gender metadata. Global outputs change in 3/105 full-transcript judgments and 2/105 participant-only judgments, for a total of 5/210 changes. Item-threshold outputs are more sensitive, changing in 14/210 pooled judgments. The largest shift occurs for participant-only prompts: adding recorded gender metadata reduces item-threshold positive judgments from 14 to 8 and lowers F1 by 0.122.

From a measurement perspective, this analysis treats recorded gender metadata as a facet perturbation: the transcript content is held fixed while the metadata field is added or omitted. A stable scoring procedure should not change substantially under this perturbation. The global binary format shows relatively small metadata sensitivity, whereas item-threshold scoring is more affected. A likely explanation is that item-threshold scoring converts an accumulated evidence total into a binary decision, so small item-score shifts can move cases across the cutoff of 10.

This result should be interpreted as a limited measurement-invariance check, not as a fairness audit. We test only whether adding recorded gender metadata changes judgments for the same transcript; we do not evaluate counterfactual gender swaps, nonbinary identities, intersectional effects, or downstream clinical consequences. The appropriate conclusion is that global judgments are more stable under this metadata perturbation in our sample, while item-threshold judgments show greater facet sensitivity.

4 Limitations

This study has several limitations. First, the DAIC-WOZ development split is small, with only 35 interviews. As a result, performance estimates, reliability coefficients, and facet-effect estimates may be sensitive to individual cases. Second, each model-condition-transcript cell is evaluated only once. Because $k = 1$, we cannot estimate repeated-call stability or separate within-model sampling noise from other sources of variation. A stronger design would use more repeated calls per cell, such as $k = 10$, but we prioritized covering more input conditions and judge models within the available time and compute budget.

Third, PHQ-8 labels are self-reported screening labels rather than clinical diagnoses. Fourth, we evaluate text-only transcripts and do not use audio or visual signals from the original clinical interview setting. Fifth, we evaluate only three LLM judge models and two output formats. Other models, prompts, temperatures, or calibration procedures may behave differently. Sixth, our gender analysis uses recorded gender metadata and does not include counterfactual gender variants, nonbinary identities, or intersectional demographic comparisons. Therefore, it should be interpreted as a metadata-sensitivity check, not as a full fairness study.

Finally, strong benchmark performance or high measurement reliability should not be interpreted as clinical readiness. Automated mental-health judgments require human oversight, privacy protections, and careful deployment constraints.

5 Conclusion and Future Work

We evaluated zero-shot LLM mental-health judges using speaker ablations, output-format comparisons, cross-model reliability analyses, Generalizability Theory, many-facet Rasch-style modeling, and recorded-gender metadata perturbations. Across participant-containing prompts, direct global binary judgments outperform item-threshold predictions on recall, F1, MCC, cross-model reliability, and dependability. Item-threshold predictions are more conservative: they produce fewer false positives, but miss many depression-positive cases and require aggregation across multiple judge models and input conditions to reach acceptable dependability.

The interviewer-only ablation provides the clearest cautionary result. Although pooled interviewer-only performance is weak, Claude’s global interviewer-only judgments remain surprisingly predictive. This partially echoes prior findings that interviewer prompts and positions can support depression classification through shortcut artifacts, while showing that zero-shot LLM shortcut sensitivity is model-specific. Recorded-gender metadata has a smaller effect on global judgments than on item-threshold outputs, but this analysis should be interpreted only as a limited metadata-sensitivity check, not as a full fairness conclusion.

Overall, our results suggest that LLM-based depression judgments should be treated as preliminary screening signals, not diagnostic decisions. Measurement-oriented evaluation should report participant-only, full-transcript, and interviewer-only ablations; compare output formats; test cross-model reliability; and check whether demographic metadata changes outputs. Future work should extend this analysis to larger datasets, add repeated-call reliability designs, test counterfactual demographic metadata, and evaluate prompting or calibration strategies that reduce reliance on interviewer-side artifacts.

6 Ethics Statement

Disclosure. To maintain scientific standards, we treated all generated text, code, and analysis suggestions as drafts requiring human verification rather than as final authority. We checked that the report used our own wording and that ideas from prior work were attributed with citations, especially for LLM-as-judge evaluation, DAIC-WOZ, PHQ-8, interviewer-side shortcuts, Generalizability Theory, and many-facet Rasch-style modeling. We avoided presenting LLM outputs as clinical diagnoses and consistently described PHQ-8 labels as screening labels. To address bias and inaccuracies, we included speaker ablations, cross-model reliability analysis, gender-metadata sensitivity checks, and measurement-dependability analyses rather than relying only on predictive performance. We also reported limitations, including the small DAIC-WOZ development split, the use of recorded gender metadata rather than counterfactual gender interventions, and the descriptive nature of the binary

G-theory and Rasch-style analyses. Final interpretations were reviewed for neutrality, caution, and consistency with the reported results.

AI Reflection. We used AI tools as support for brainstorming, statistical interpretation, and writing revision. They helped us clarify measurement concepts such as inter-rater reliability, Generalizability Theory, dependability coefficients, D-studies, and many-facet Rasch-style models. AI tools were especially useful for translating abstract course concepts into concrete project analyses, including speaker ablations, cross-model reliability checks, metadata-sensitivity tests, and G-theory dependability tables. They also helped improve the clarity of LaTeX tables, section structure, and academic phrasing. However, we treated AI outputs as drafts rather than final authority. We checked formulas, numerical interpretations, citations, and claims against our own data, code, and course materials. The main learning benefit was that AI tools made it easier to iterate between conceptual understanding and implementation. In the future, we would use AI similarly: as a research assistant for explanation and organization, while retaining human responsibility for verification, attribution, and final scientific judgment.

Impact. This work studies LLM-based depression judgments from clinical interview transcripts, so the potential social and ethical impacts are significant. Such systems could help researchers audit measurement reliability or explore preliminary screening signals, but they could also be misused as stand-alone clinical decision makers. Misuse could lead to false reassurance, unnecessary concern, demographic bias, or overreliance on model outputs without professional evaluation. To mitigate these risks, we frame the task as measurement analysis rather than diagnosis, emphasize PHQ-8 labels as screening labels, and recommend input ablations, cross-model reliability checks, and metadata sensitivity tests before any high-stakes use. We use the DAIC-WOZ development split only for analysis and report aggregate results rather than exposing participant identities. We also distinguish limited gender-metadata sensitivity from a full fairness audit.

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